



Teheran University of Medical Sciences

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Final Exam
International Campus, DDS / Pharma D./MD Program
Biochemistry
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Student ID Number:

Course Coordinator: Dr. Nourozi

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Duration of Exam: 50 Minutes

MCQ Question	50
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



Which of the following is synthesized In Phosphogluconate pathway :		1
NADH	الف	
FADH	ب	
NADPH	ج	
GTP	د	

Which of the following is a high energy molecule?		2
dihydroxyacetone-P	الف	
glyceraldehydes 3- P	ب	
1,3-biphosphoglycerate	ج	
2 phosphoglycerate	د	

Which of the following enzymes is important in fermentation?		3
LDH	الف	
ALT	ب	
AST	ج	
ALP	د	

Which of the following enzymes is important in glycogenolysis?		4
Phosphorylase	الف	
Hexokinas	ب	
Glucokinase	ج	
Phosphatase	د	

Which glucose transporter is active in liver tissues?		5
GLUT-1	الف	
GLUT-2	ب	
GLUT-3	ج	
GLUT-4	د	

All the following enzymes are important in gluconeogenesis except:		6
PEPCK	الف	
Fructose 1,6 biphosphatase	ب	
Glucose 6 phosphatase	ج	
Phosphorylase	د	

How many pyruvate molecules are required for gluconeogenesis?		7
1	الف	
2	ب	
3	ج	
4	د	

Where does glycolysis take place?		8
Cytosol	الف	
Mitochondria	ب	
Nucleus	ج	
Cell membrane	د	

Oxidation of one glucose molecule yields how many ATP molecules?		9
26-28	الف	
30-32	ب	
34-36	ج	
38-40	د	

For cellulose synthesis which molecule is important in glucose activation?		10
ADP	الف	
CDP	ب	
GDP	ج	
TDP	د	



Which molecule is synthesized in HMP shunt?		11
NAD	الف	
NADH	ب	
FADH	ج	
NADPH	د	

How many ATP molecules are synthesized in glycolysis?		12
1	الف	
2	ب	
3	ج	
4	د	

HMG-CoA reductase activity is increased by the administration of the hormone:		13
Insulin	الف	
Epinephrine	ب	
Glucagon	ج	
Glucocorticoids	د	

Anti-inflammatory corticosteroids inhibit...		14
Phospholipase A1	الف	
Lipo-oxygenase	ب	
Phospholipase A2	ج	
Cyclo-oxygenase	د	

Genetic deficiency of lipoprotein lipase causes hyperlipoproteinemia of the following type:		15
Type I	الف	
Type IIa	ب	
Type IIb	ج	
Type V	د	



Acetyl CoA formed from pyruvate can be used for the synthesis of all the following except:		16
Cholesterol	الف	
Fatty acids	ب	
Steroid hormones	ج	
Glucose	د ✓	

NADPH is produced when this enzyme acts.		17
Pyruvate dehydrogenase	الف ✓	
Malic enzyme	ب	
Succinate dehydrogenase	ج	
Malate dehydrogenase	د	

Long-chain fatty acyl CoA esters are transported across the mitochondrial membrane by.....		18
cAMP	الف	
Choline	ب ✓	
Prostaglandin	ج	
Carnitine	د ✓	

During each cycle of β-oxidation of fatty acid, all the following compounds are generated except:		19
NADH	الف	
H ₂ O	ب	
FADH	ج	
Acyl CoA	د ✓	

Acetyl CoA required for fatty acid synthesis is produced by:		20
Pyruvate dehydrogenase complex	الف	
Citrate lyase	ب	
Thiolase	ج ✓	
Carnitine-acyl transferase	د	



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All statements regarding 3-OH-3 methyl glutaryl CoA are true except:		21
It is formed in the cytoplasm	الف	
Required in ketogenesis	ب	
Involved in the synthesis of Fatty acid	ج	
An intermediate in cholesterol biosynthesis	د	

Which of the following is an activator of fatty acid synthesis?		22
Palmitate	الف	
Carnitine	ب	
Malonyl CoA	ج	
Citrate	د	

1 molecule of palmitic acid in total oxidation to Co₂ will yield molecules of ATP (as high energy bonds):		23
119	الف	
129	ب	
146	ج	
154	د	

β-Oxidation of odd-carbon fatty acid chain produces:		24
Propionyl CoA	الف	
Malonyl CoA	ب	
Succinyl CoA	ج	
Glutaryl CoA	د	

In the respiratory chain, cytochrome a₃ is located in complex.....		25
I	الف	
II	ب	
III	ج	
IV	د	

Which part of the respiratory chain has the lowest oxidizing power?			26
FMN	الف		
Coenzyme Q	ب		
Cytc	ج		
Cytc	د		

The ATP/ADP transporter in the mitochondrial inner membrane is inhibited by			27
Oligomycin	الف		
CN ⁻	ب		
Attractyloside	ج		
Barbiturates	د		

Which of the complexes of the respiratory chain does not act as a proton pump?			28
Succinate Q Reductase	الف		
NADH dehydrogenase	ب		
Cytochrome C oxidase	ج		
Cyt C reductase	د		

Deficiency or hereditary absence of which enzyme causes immunodeficiency in infants?			29
Thymidylate synthetase	الف		
Adenosine deaminase	ب		
Xanthine oxidase	ج		
Ribonucleotide reductase	د		

All the following compounds are involved in the biosynthesis of pyrimidine nucleotides except:			30
Aspartate	الف		
Phosphoribosyl pyrophosphate	ب		
Carbamyl phosphate	ج		
Glycine	د		



In humans the end product of purine catabolism is...		31
Urea	الف	
Xanthine	ب	
Uric acid	ج	
Allantoin	د	

Lesch-Nyhan syndrome, the sex-linked recessive disorder is due to the lack of the enzyme:		32
Hypoxanthine-guanine phosphoribosyl transferase	الف	
Adenosine deaminase	ب	
Xanthine oxidase	ج	
Adenine phosphoribosyl transferase	د	

In which of following hyperbilirubinemias (jaundices), the amount of <u>conjugated bilirubin</u> in the blood would increase the most?		33
Obstructive	الف	
Hepatitis	ب	
Hemolytic	ج	
Physiologic	د	

The first nucleotide formed during purine de novo synthetic pathway is?		34
PRPP	الف	
Uric acid	ب	
OMP	ج	
IMP	د	

Which enzyme is inhibited by Allopurinol to treat which disease?		35
Orotate decarboxylase, Renal Azotemia	الف	
Xanthine oxidase; Gout	ب	
Orotate decarboxylase, Gout	ج	
Xanthine oxidase; SCID	د	



..... donates its phosphate to ADP to yield ATP during muscle exertion.		36
Arginine glycine amidinotransferase	الف	
Creatine phosphate	ب	
Creatine kinase	ج	
Guanidoacetate	د	

All the following are nonfunctional enzymes <u>except</u>:		37
Alkaline phosphatase	الف	
Acid phosphatase	ب	
Lipoprotein lipase	ج	
Gamma glutamyltranspeptidase	د	

Which among the following is <u>not</u> a pancreatic specific enzyme?		38
Lipase	الف	
Aldolase	ب	
Trypsin	ج	
Amylase	د	

Which of the following is <u>the only</u> plasma enzyme normally found in urine?		39
Acid Phosphatase	الف	
Nucleotidase	ب	
Nucleotidase	ج	
Amylase	د	

Which among the following is <u>not</u> a pancreatic specific enzyme?		40
Lipase	الف	
Aldolase	ب	
Trypsin	ج	
Amylase	د	



Which of the following statements about the urea cycle is correct?		41
Argininosuccinate is lysed to urea and ornithine in the urea cycle.	الف	
Carbamoyl phosphate supplies both of the nitrogen atoms of urea in the urea cycle.	ب	
Arginine is hydrolysed to urea and ornithine in the urea cycle.	ج	
The formation of urea from the urea cycle yields energy.	د	

Which of the following statements about transamination reactions is correct?		42
Transamination reactions involve ATP hydrolysis.	الف	
Transamination reactions are irreversible.	ب	
Transamination reactions require NAD^+ or NADP^+ .	ج	
Transamination reactions require pyridoxal-5'-phosphate.	د	

Which of the following amino acids <u>cannot</u> provide a substrate for gluconeogenesis?		43
Leucine	الف	
Tryptophan	ب	
Histidine	ج	
Isoleucine	د	

Which of the following is non-essential amino acid?		44
Lysine	الف	
Serine	ب	
Leucine	ج	
Threonine	د	

The amino acid which serves as a carrier of ammonia from skeletal muscle to liver is:		45
Alanine	الف	
Methionine	ب	
Arginine	ج	
Glutamine	د	

COUNTINUE TO THE NEXT PAGE....



Amino acids are released into the portal system by:		46
Pinocytosis	الف	
Active transport	ب	
Facilitated transport	ج	
Passive transport	د	

Which of the following enzyme catalyze the major mechanism for transporting ammonia in a nontoxic form?		47
Aminotransferase	الف	
Amino acid oxidase	ب	
Glutamine synthetase	ج	
Threonine dehydrogenase	د	

Which of the following enzyme is deficient in alkaptonuria?		48
Homogentisate dioxygenase	الف	
p-Hydroxyphenyl- pyruvate dioxygenase	ب	
Phenylalanine hydroxylase	ج	
Tyrosinase	د	

A person with phenylketonuria cannot convert ...		49
Phenylalanine to tyrosine	الف	
Phenylalanine to isoleucine	ب	
Phenol into ketones	ج	
Phenylalanine to lysine	د	

An example of a transamination process is		50
glutamate = hexanoic acid + NH ₃	الف	
aspartate + hexanoic acid = glutamate + oxaloacetate	ب	
aspartate + α ketoglutarate = glutamate + oxaloacetate	ج	
glutamate = α-ketoglutarate + NH ₃	د	